

Baseball Pitch Prediction

The Idea

Build a machine-learning system that predicts both:

1. **What type of pitch an MLB pitcher will throw next**
2. **Where that pitch will be thrown**

These will be treated as two separate prediction problems:

- **Pitch Type Model:** predicts fastball, slider, curveball, changeup, etc.
- **Pitch Zone Model:** predicts which region of the strike zone the pitch will target.

The main goal is to determine:

How accurately can we predict both pitch selection and pitch location using game context, pitcher tendencies, and recent pitch history?

MVP

By the end of the term, the system should:

1. Build a clean pitch-by-pitch MLB dataset.
2. Establish simple prediction baselines.
3. Train an XGBoost model for **pitch type**.
4. Train a separate XGBoost model for **pitch zone**.
5. Analyze which features contribute most to each prediction task, and compare different feature groups through ablation experiments.
6. Build an interactive demo displaying both predictions.

For pitch type, we would likely classify pitches into:

- Four-seam fastball
- Sinker
- Slider
- Curveball
- Changeup
- Cutter
- Other

For pitch location, we would divide the strike zone into regions and treat the problem as multiclass classification. We can start with a 3×3 grid and potentially add regions for pitches outside the strike zone.

Simple baselines could include:

- Pitcher's most common pitch / zone
- Pitcher's most common pitch / zone given the current count

Data Source

We will use **MLB Statcast data from Baseball Savant**, which provides pitch-level information including pitch type, location, count, game state, batter/pitcher handedness, velocity, movement, spin rate, and pitch results.

We can retrieve the data programmatically using the **pybaseball** (Statcast API), allowing us to collect multiple MLB seasons and build our own chronological training dataset.

XGBoost Models and Features

We will train two independent XGBoost models using largely the same feature set:

- **XGBoost #1 → Pitch Type**
- **XGBoost #2 → Pitch Zone**

Features:

Current Game Situation:

- Balls
- Strikes
- Outs
- Inning
- Top / bottom of inning
- Score differential
- Runners on first, second, and third
- Number of runners on base
- Batter handedness
- Pitcher handedness

Pitcher's overall tendencies:

- Fastball usage %
- Slider usage %
- Curveball usage %
- Changeup usage %
- Sinker usage %
- Cutter usage %

- Other pitch usage %

Pitcher tendencies by batter handedness:

- Pitch-type usage vs left-handed batters
- Pitch-type usage vs right-handed batters

Pitcher tendencies by count:

- Pitch-type usage for the current count
- Pitch-type usage when ahead in the count
- Pitch-type usage when behind in the count
- Pitch-type usage with two strikes
- Pitch-type usage on full counts

Previous pitch types:

- Previous pitch type
- Pitch type 2 pitches ago
- Pitch type 3 pitches ago
- Potentially previous 4–5 pitches

Previous pitch locations:

- Previous pitch zone
- Zone 2 pitches ago
- Zone 3 pitches ago

Previous pitch results:

- Ball
- Called strike
- Swinging strike
- Foul
- Ball in play
- Whether the batter swung

Previous pitch characteristics:

- Velocity
- Spin rate
- Movement
- Release location

Current at-bat history:

- Number of pitches in the plate appearance
- Number of fastballs thrown

- Number of breaking balls thrown
- Number of off-speed pitches thrown
- Number of consecutive pitches of the same type

Ablation experiments: We can perform multiple trainings using different feature groups to measure how much game context, pitcher tendencies, and pitch history each improve performance.

If We Have Time — Sequence Modeling

If the XGBoost models are completed early, we can experiment with a sequence model that directly processes the ordered sequence of previous pitches rather than representing them through fixed features.

Possible approaches:

- **LSTM** — standard recurrent sequence model
- **GRU** — simpler/faster recurrent alternative
- **Transformer** — more complex model capable of learning relationships across longer pitch sequences

Each pitch in the sequence could contain information such as:

- Pitch type
- Pitch zone
- Result
- Count
- Velocity

The main comparison would be:

Does explicitly modeling pitch sequences outperform XGBoost using engineered pitch-history features?

Depending on time, this could be tested for pitch type, pitch zone, or both.

Demo

Build a simple interactive dashboard where a user can select a pitcher and enter the current game state / previous pitch sequence.

The dashboard would display:

- Predicted probabilities for the next pitch type

- Predicted probabilities over the strike-zone regions
- A visual heatmap of the predicted pitch location
- Relevant pitcher tendencies and recent pitch history
- A local explanation of the prediction, showing which features, most influenced the pitch-type and pitch-zone outputs

For example, the explanation component could highlight which features pushed the model toward a specific pitch type or zone, using feature contribution methods such as XGBoost feature importance or SHAP-style explanations.

The demo could also replay historical plate appearances pitch-by-pitch, showing both the model's prediction and the features that most influenced that prediction before each actual pitch.

Timeline

Weeks 1–2 — Data

- Collect and clean pitch-level data
- Define pitch categories and strike-zone regions
- Build the feature-generation pipeline
- Create train/validation/test datasets

Weeks 3–4 — Baselines

- Implement pitch-type baselines
- Implement pitch-zone baselines
- Perform initial exploratory analysis

Weeks 5–7 — XGBoost

- Train pitch-type XGBoost model
- Train pitch-zone XGBoost model
- Tune models
- Run feature-group / ablation experiments

Weeks 8–9 — Analysis

- Feature importance
- Performance by pitcher/count/pitch type
- Compare which features matter for type vs location

Weeks 10–11 — Final Product / Sequence Model

- Build interactive demo
- If time permits, train an LSTM, GRU, or Transformer and compare against XGBoost

Week 12 — Final Deliverable

- Final models
- Analysis and visualizations
- Interactive demo
- GitHub repository
- Final presentation / report

(Bonus Section) Why XGBoost / Why Classification?

Why XGBoost?

XGBoost is a strong fit because most of our inputs are structured/tabular features, and many of the relationships between them are nonlinear. For example, the effect of count may depend on batter handedness, pitcher tendencies, and previous pitches.

XGBoost also gives us:

- Strong performance on tabular data
- Ability to capture feature interactions without manually defining them
- Relatively fast training and iteration
- Feature importance / SHAP explanations for the demo

Why classify pitch zones instead of regress exact location?

Although Statcast provides exact `plate_x` and `plate_z` coordinates, we plan to convert these into discrete strike-zone regions and treat location as a classification problem.

Classification is preferable for the MVP because:

- The prediction is easier to interpret e.g. **low-and-away** rather than (0.73, 1.91)
- It naturally provides probabilities across possible locations
- Small coordinate errors do not necessarily represent meaningful baseball differences
- It makes evaluation and visualization in the demo straightforward